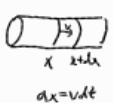


el. tok v raztopini elektrolitov



shovni tok

$$J_i = \frac{dn_i}{dt} = c_i \frac{dV}{dt} = c_i S v_i$$

$$dx = v_i dt$$

el. tok

$$I_i = J_i z_i F = S F z_i c_i v_i$$

↳ Faradayeva konst.

gostota el. toka

$$i_i = \frac{I_i}{S} = F z_i c_i v_i$$

$$i = \sum_i F z_i c_i v_i$$

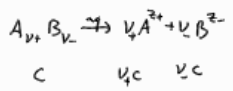


$$I = \frac{U}{R} = \frac{U S}{l} = S F \sum_i z_i c_i v_i \quad / : \frac{U}{l} = E$$

$$\kappa = \frac{F}{E} \sum_i z_i c_i v_i = F \sum_i z_i c_i \underbrace{\frac{v_i}{E}}_{\text{gibljivost } u_i}$$

$$R = \sum \frac{l}{S} = \frac{1}{\kappa} \frac{l}{S}$$

↳ spec. prevodnost



$$\Lambda = \frac{\kappa}{c} = F \sum_i z_i \frac{c_i}{c} u_i = F \sum_i z_i v_i u_i$$

↳ molska prevodnost

1 elektrolit

$$\Lambda = v_+ F z_+ u_+ + v_- F z_- u_- = v_+ \lambda_+ + v_- \lambda_-$$

↳ ionska prevodnost

Kohlrausch:  $\lambda \text{ [cm}^2 \Omega^{-1} \text{ mol}^{-1}]$

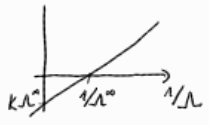
$$\Lambda^\infty = v_+ \lambda_+^\infty + v_- \lambda_-^\infty$$

neskončno razredjenje (ioni se ne zadržijo med sabo)

močni elektroliti:  $\Lambda = \Lambda^\infty - A \sqrt{c}$



šibki elektroliti:  $\Lambda \frac{c}{c^\circ} = K (\Lambda^\infty)^2 \frac{1}{\Lambda} - K \Lambda^\infty$



Transportno število

$$t_i = \frac{i_i}{i}, \text{ 1 elektrolit}$$

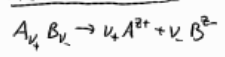
$$t_+ = \frac{i_+}{i} = \frac{F c_+ z_+ u_+}{F c_+ z_+ u_+ + F c_- z_- u_-} = \frac{c_+ \lambda_+}{c_+ \lambda_+ + c_- \lambda_-}$$

elektro neutralnost:  $c_+ z_+ = |c_- z_-|$

$$\rightarrow t_+ = \frac{u_+}{u_+ + u_-}, \quad t_- = \frac{u_-}{u_+ + u_-}$$

$$t_+ + t_- = 1$$

Aktivnost elektrolita



$$\begin{matrix} n_2 & v_+ n_2 & v_- n_2 \\ dn_2 & v_+ dn_2 & v_- dn_2 \\ & || & || \\ & dn_+ & dn_- \end{matrix}$$

$$dG = -SdT + Vdp + \sum \mu_i dn_i$$

$$\left( \frac{\partial G}{\partial p} \right)_{T,p_i} = v_+ dn_+ + v_- dn_- \stackrel{\text{iz var. minimuma}}{=} 0$$

$$\mu_i = \mu_i^\circ + RT \ln a_i = \mu_i^\circ + RT \ln \left( \gamma_i \frac{c_i}{c^\circ} \right)$$

$$\mu_2 = \left( \frac{\partial G}{\partial n_2} \right)_{T,p,n_1} = v_+ \mu_+ + v_- \mu_-$$

$$= v_+ \left( \mu_+^\circ + RT \ln \gamma_+ \frac{c_+}{c^\circ} \right) + v_- \left( \mu_-^\circ + RT \ln \gamma_- \frac{c_-}{c^\circ} \right)$$

$$= \underbrace{v_+ \mu_+^\circ + v_- \mu_-^\circ}_{\mu_2^\circ} + RT \ln \left( \gamma_+ \frac{c_+}{c^\circ} \right)^{v_+} \left( \gamma_- \frac{c_-}{c^\circ} \right)^{v_-}$$

$c_i = v_+ c, \quad v = v_+ + v_-$

$$\mu_2 = \mu_2^\circ + RT \ln a_2$$

$$a_2 = \gamma_+^{v_+} \gamma_-^{v_-} \left( \frac{c}{c^\circ} \right)^v = a_\pm^v \rightarrow a_\pm = \left( \gamma_+^{v_+} \gamma_-^{v_-} \right)^{1/v} \left( \frac{c}{c^\circ} \right)^{1/v}$$

↳ srednji aktivnostni koef.

$$\gamma_\pm = \gamma_+^{v_+} \gamma_-^{v_-} \rightarrow \gamma_\pm = (\gamma_+^{v_+} \gamma_-^{v_-})^{1/v}$$

↳ srednji koeficient aktivnosti

## Elektrokemijski člani

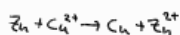
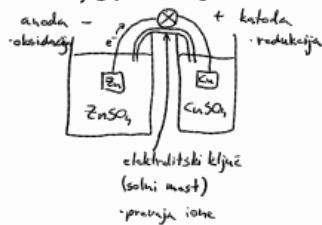
### 1) Elektrolizni člani

$$q = z F n = \int j dt$$

· proizvodi  $H_2, O_2, Al$

· za proces moramo dovesti E

### 2) Galvanski člani



$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q_a =$$

$$= \Delta_r G^\ominus + RT \ln \frac{a(Cu) a(Zn^{2+})}{a(Zn) a(Cu^{2+})}$$

spontana reakcija:  $\Delta_r G < 0$

$$\Delta_r G \leq W_{\text{max}}$$

||

$$W_{\text{el}} = -zFE$$

$$-zFE = -zFE^\ominus + RT \ln Q_a$$

$$E = E^\ominus - \frac{RT}{zF} \ln Q_a$$

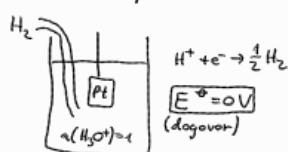
Daniellov član:  $E^\ominus = E_{Cu^{2+}/Cu}^\ominus - E_{Zn^{2+}/Zn}^\ominus$ ,  $a(Zn) = a(Cu) = 1$

$$E_{Cu/Zn} = E_{Cu^{2+}/Cu}^\ominus - E_{Zn^{2+}/Zn}^\ominus - \frac{RT}{zF} \ln \frac{a(Zn^{2+})}{a(Cu^{2+})} =$$

$$= E_{Cu^{2+}/Cu}^\ominus + \frac{RT}{zF} \ln a(Cu^{2+}) - \left( E_{Zn^{2+}/Zn}^\ominus + \frac{RT}{zF} \ln a(Zn^{2+}) \right) =$$

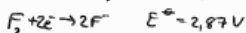
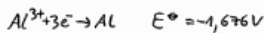
$$= E_{Cu} - E_{Zn}$$

Standardni H-polčlan



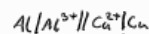
$$E(H_2) = 0 + RT \ln \frac{(p/p^\ominus)^{1/2}}{a(H^+)}$$

Redox vrsta



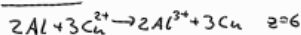
Primer:

izračun napetosti člana



$T = 25^\circ C$  → pokrajna črta predstavlja fazno mejo (raztopina-trdno/razt.-gel)

$$a_i = 0,1$$

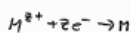


$$E = E_{Cu}^\ominus - E_{Al}^\ominus$$

$$E = E^\ominus - \frac{RT}{zF} \ln \frac{a^3(Cu) a^2(Al^{3+})}{a^3(Al) a^3(Cu^{2+})} = 2,008 V$$

Tipi elektrod

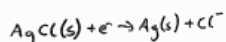
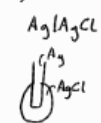
### 1) I. red



· elektroda neposredno sodeluje v reakciji

$$E = E_{M^{z+}/M}^\ominus + \frac{RT}{zF} \ln a(M^{z+})$$

2) II. red



$E_{\text{AgCl}} = E_{\text{Ag}^+/\text{Ag}}^\circ + \frac{RT}{F} \ln \frac{a(\text{Ag}^+) a(\text{Cl}^-)}{a(\text{AgCl})}, K_{sp} = a(\text{Ag}^+) a(\text{Cl}^-)$

$E_{\text{AgCl}} = E_{\text{Ag}^+/\text{Ag}}^\circ + \frac{RT}{F} \ln \frac{K_{sp}}{a(\text{Cl}^-)}$

$E_{\text{Ag}} = E_{\text{AgCl}/\text{Ag}}^\circ - \frac{RT}{F} \ln a(\text{Cl}^-)$

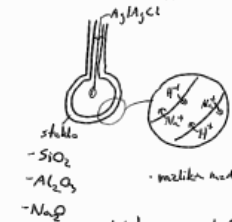
3) Redoks elektroda



4) Membranska elektroda

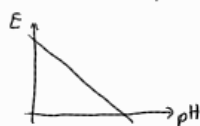
• steklena elektroda

+ referenčna elektroda



• mešični acid prehodi  $\text{Na}^+$  in  $\text{H}^+$  se izrazi kot napetost

• lahko merimo tudi pH:



$E_{st} = E_{st}^\circ - \frac{2303RT}{F} \text{pH}$